CLOUD COMPUTING

How online storage and service work

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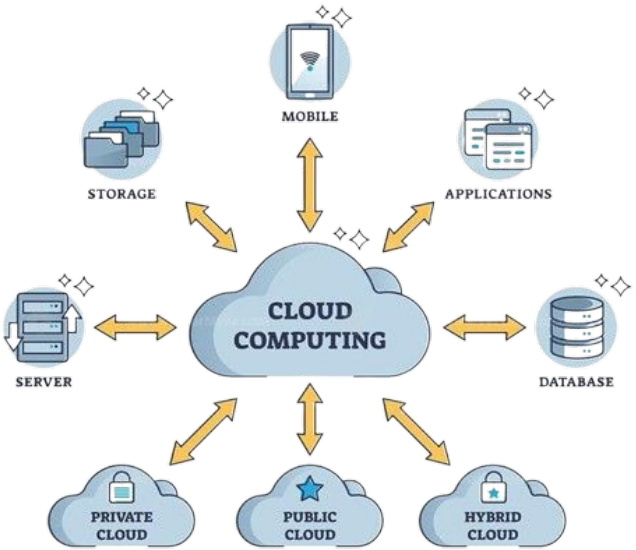
What is Cloud Computing?

Cloud computing provides on-demand access to computing resources-such as servers, storage, applications, and databases-over the internet, without requiring users to own and maintain the underlying infrastructure.



Background of Cloud Computing?

- Cloud computing is a technology that enables users and organizations to access computing resources over the internet without having to maintain all the necessary physical infrastructure locally.
- It provides on-demand access to servers, storage, databases, networking, and software. This approach improves scalability, flexibility, and resource utilization while reducing the costs associated with purchasing and maintaining hardware.
- Cloud computing has become an essential part of modern information technology, supporting digital services, business operations, data management, and application development.



How it Works?







Your Device

You send a request(e.g. Open a file.)

The Internet

Your request travels securely over the internet.

Data Center

Services and data are stored on powerful remote servers.

The service stores and returns information when you need it.

Cloud Service Models

SaaS	**Software as a Service** Use an app online, such as Google Drive.	
PaaS	**Platform as a Service** A place with tools for developers to build apps.	
IaaS	**Infrastructure as a Service.** Rent computing resources such as servers and storage	

Daily Life Examples



Google Drive

Stores and access file from anywhere



Streaming

Watch videos or listen to music online



Online Class

Join and access course material online.



Photo Backup

Keep your photos safe in the cloud

Benefits of Cloud Computing

Cloud services can make everyday work easier to access and share

1

Access Across Devices

Open your work from different connected devices.

2

Easy Collaboration

Share a file so team can view or edit it.

3

Flexible Storage

Keeping files online without storing every copy on device.

Challenges and Risks



Internet Outage

Hard to access files and services without internet connection.



Privacy Concerns

Your data is stored on external servers and could be targeted by attackers.



Accidental Sharing

Files might be shared with the wrong people by mistake.

Staying Safe in the Cloud



Use Strong Passwords

Create unique, complex passwords for your account.



Enable Two-Step Verification

Add an extra layer of security when possible.



Check Sharing Permissions

Make sure you’re only sharing files with the right people.

Conclusion

Cloud computing makes it easier to access, store, and share data and services online. It offers flexibility for people and organizations wherever they have an internet connection.

Key Takeaway

- Access files from anywhere with an internet connection.
- Store and share data online instead of keeping everything on one device.
- Use apps and services when needed without installing everything locally.
- Scale resources up or down as needs change.

References and Sources

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